

Hardware procurement and reuse strategy

A staged, 24 VDC-first component plan for Projects 1-4. Buy the shared bench once, then add only the process hardware required by each project.

SAFETY BOUNDARY

The required portfolio can be completed with extra-low-voltage 24 VDC equipment. Do not expose mains terminals, energize a three-phase motor/VFD, build a safety-rated circuit, or pressurize a process unless a competent person has designed, installed and supervised that work. A mushroom pushbutton connected only to a standard PLC is a training input - it is not a validated emergency-stop safety function.

System architecture

Layer	Baseline decision
Power	Certified external 24 VDC supply or professionally enclosed DIN-rail supply; individually fused branches.
Control	Siemens S7-1200 DC/DC/DC PLC; add a compatible analog module for 4-20 mA work.
Field layer	Standardize on 24 VDC, 3-wire PNP sensors where practical; document every pinout.
Actuation	Use interposing relays or correctly rated low-voltage drivers; add suppression across DC coils.
Supervision	Laptop with TIA Portal/PLCSIM and Ignition; a physical HMI is optional.
Evidence	Labelled terminal plan, I/O list, schematics, photos, test sheets, fault log and backups.

Buying rules

- [] Confirm the exact PLC article number, generation and supported TIA Portal version before paying.
- [] Check voltage, output type, load current, inrush current, sensor wiring and common/reference arrangement.
- [] Choose spare I/O and terminal capacity; do not size the system to the final channel.
- [] Buy one representative device first, bench-test it, then duplicate only after the interface is proven.
- [] Record supplier, article number, datasheet revision, warranty and measured commissioning result.

RECOMMENDED CONTROLLER BASELINE

Example: SIMATIC S7-1200 CPU 1212C DC/DC/DC, 6ES7212-1AE40-0XB0 (8 digital inputs, 6 transistor outputs and 2 onboard 0-10 V analog inputs), plus SM 1231 AI, 6ES7231-4HD32-0XB0 for four voltage/current inputs including 4-20 mA. Treat these as a coherent example, not an instruction to mix generations or incompatible modules.

Master reusable component list - control bench

Required quantities are the practical minimum for one learner bench. Alternatives are acceptable when they meet the stated interface and safety requirements.

Component	Qty	Minimum specification	Reuse	Procurement note
PLC CPU	1	24 VDC Siemens S7-1200 DC/DC/DC; Ethernet; at least 8 DI, 6 DO, 2 AI	P1-P4	Buy first; verify TIA compatibility
Analog input	1	4-channel differential voltage/current; supports 0/4-20 mA	P2-P4	Required for real transmitters
24 VDC supply	1	Certified, regulated, current-limited; target at least 4 A after load calculation	P1-P4	External enclosed adapter is safest for solo work
DC protection	1 set	Branch fuses or electronic protection; disconnect; labelled distribution	P1-P4	Protect PLC, field and actuator branches separately
Ethernet switch	1	Unmanaged 5-port, 24 VDC industrial or reputable bench unit	P1-P4	PLC, laptop, SCADA and future device
DIN rail / base	1	Stable, ventilated baseboard or enclosure; wire duct optional	P1-P4	No exposed mains terminals
Terminal blocks	30+	Feed-through, fused, disconnect/test and protective-earth types as needed	P1-P4	Add 20% spare positions
Interposing relays	4	24 VDC coil, contacts rated for the actual load; LED/test lever useful	P1-P4	Use suppression; never exceed contact rating
Pushbuttons	4	22 mm momentary: Start, Stop, Reset and acknowledge/jog	P1-P4	At least one NC Stop contact
Selector switches	2	Maintained 2/3-position: Auto/Manual and Local/Remote	P1-P4	Label positions clearly
Pilot lights	4	24 VDC red, amber, green and blue/white	P1-P4	Power, run, warning and fault
E-stop device	1	Dual-channel mushroom device for training/interface demonstration	P1-P4	Not safety-rated when wired only to standard PLC

POWER-SUPPLY EXAMPLE

A reputable DIN-rail example is Mean Well HDR-100-24N (24 V, 4.2 A), but its mains input must be enclosed and installed by a competent person. Recalculate capacity from measured loads; the model name is not a substitute for a power budget.

Master reusable component list - field, tools and evidence

Component	Qty	Minimum specification	Reuse	Use note
Photoelectric sensors	2	24 VDC PNP, light/dark selectable if possible	P1, P4	Object detection and injected faults
Inductive sensor	1	24 VDC PNP, normally open, suitable target distance	P1, P4	Metal target/proximity demonstration
Limit switch	2	Mechanical roller/plunger, one changeover contact	P1, P2, P4	Position and overtravel scenarios
Analog simulator	1	0-10 V potentiometer module or protected 10 kOhm potentiometer	P1-P4	Commission logic before a transmitter
4-20 mA source	1	Loop simulator/source with clear source/sink modes	P2-P4	Prefer fused/protected terminals
Digital multimeter	1	DC V, resistance, continuity and DC mA with fused current input	P1-P4	Know lead position before every current test
Wiring kit	1 set	0.5-0.75 mm ² control wire, ferrules, jumpers, cable markers, glands	P1-P4	Colour-code consistently
Hand tools	1 set	Ferrule crimper, stripper, cutters and insulated screwdrivers	P1-P4	Use correct ferrule/crimp size
Patch/test leads	1 set	Shrouded leads and labelled low-voltage patch leads	P1-P4	Avoid improvised exposed conductors
Spare devices	1 set	One relay, fuse set, sensor and terminal jumpers	P4	Supports substitution troubleshooting
Laptop	1	Ethernet/USB, TIA Portal/PLCSIM-compatible; able to run Ignition	P1-P4	Back up software and licences
Evidence storage	1	Versioned local folder plus second backup location	P1-P4	Source, PDFs, photos, test results and exports

Optional purchases - only after a demonstrated need

Item	Decision
Physical HMI	Optional; Ignition Perspective on the laptop can provide the required interface and evidence.
Real VFD + three-phase motor	Not required. Use simulated ready/run/speed/fault I/O unless supervised by a competent person.
Safety relay	Only if completing a separately designed and validated safety circuit under competent supervision.
24 VDC UPS/buffer	Useful for Project 4 power-event testing, but simulated power-quality evidence is acceptable.
HART communicator/modem	Not required; document HART awareness and use 4-20 mA bench calibration evidence.

Project 1 component deployment

Industrial Motor & Conveyor Control Panel | Build window: 5 October-15 November 2026

Group	Quantity / specification	Reuse	Project role
Core bench	PLC, PSU, DC protection, switch, terminals, relays, buttons, selectors, lamps, training E-stop	Shared	Build and label the permanent control station
Geared DC motor	1 x 24 VDC, low-current geared motor with guarded coupling	P1, P4	Choose speed/torque for a small trainer
Motor interface	1 x rated DC motor driver or relay/contacter interface with suppression	P1, P4	PLC output must not drive motor directly
Conveyor mock-up	1 small belt/roller rig or guarded rotating target	P1	A stationary sensor target board is acceptable
Photoelectric sensors	2 x 24 VDC PNP	P1, P4	Entry and exit/object tracking
Inductive sensor	1 x 24 VDC PNP	P1, P4	Speed/proximity or metal-part test
Limit switch	1 x changeover	P1, P4	Guard/position permissive simulation
VFD interface simulator	Potentiometer, switches and lamps representing ready/run/fault/speed reference	P1, P4	Required baseline instead of a live VFD
Loads	2-4 x 24 VDC lamps or small protected loads	P1-P4	Prove output mapping before actuators

Commissioning sequence

- [] Cold checks: verify terminal numbers, polarity, continuity, fuses, relay suppression and protective separation.
- [] Power PLC only; prove input LEDs and force no outputs. Then test pilot-light loads before connecting the motor.
- [] Prove Start/Stop, controlled restart, Auto/Manual, Local/Remote, jam timer and fault reset.
- [] Inject stuck sensor, missing object, motor feedback loss, VFD-fault simulation and power-cycle recovery.
- [] Capture as-built schematic, I/O list, normal/fault/recovery results and a short narrated demonstration.

DO NOT BUY BY DEFAULT

A real VFD and three-phase motor add cost and electrical risk without being necessary to prove PLC sequencing, interfaces or troubleshooting. The simulated VFD interface is the required portfolio baseline.

Project 2 component deployment

Automated Tank, Pump & Instrumentation Skid | Build window: 16 November-27 December 2026

Group	Quantity / specification	Reuse	Project role
Core bench	PLC, analog module, PSU, protection, relays, terminals, selectors and lamps	Shared	Reuse the Project 1 control bench
Reservoirs	2 x clear, stable, low-volume plastic tanks with lids	P2, P4	Source and process/return tank
Pumps	2 x low-voltage 24 VDC pumps	P2, P4	Duty/standby or fill/transfer service
Pump interfaces	2 x rated relay/driver channels with suppression	P2, P4	Separate motor current from PLC outputs
Tubing/fittings	1 set: tubing, clamps, tees, manual valves and check valves	P2, P4	Match material and size; use drip tray
Level switches	2 x low-voltage float switches	P2, P4	Low-low and high-high independent protection
Level transmitter	1 x 4-20 mA, water-compatible, suitable range	P2-P4	May begin with signal simulator
Pressure transmitter	1 x 4-20 mA, low-pressure, water-compatible	P2-P4	Range must suit pump shutoff pressure
Flow device	1 x pulse or 4-20 mA low-flow water sensor	P2-P4	Document K-factor/range and scaling
Control valve	1 x 24 VDC solenoid or low-voltage motorized valve	P2, P4	Use rated driver and flyback suppression
Containment	Drip tray, absorbent material and cable elevation	P2, P4	Keep water below and away from electronics

Commissioning and safe-process checks

- [] Perform a dry I/O simulation first; then leak-test unpowered with electronics physically separated.
- [] Confirm transmitter loop wiring and scaling at 4, 12 and 20 mA before connecting wet instrumentation.
- [] Prove low-low trip, high-high trip, duty/standby changeover, valve fail state and restart permissives.
- [] Record pump current, fill/empty time and sensor repeatability; never exceed a device's pressure rating.

COST-CONTROL OPTION

Buy one real 4-20 mA transmitter first and use the loop simulator for the other process variables. Add a second transmitter only after the first loop is correctly wired, scaled and calibrated.

Project 3 component deployment

SCADA/HMI, Historian & Industrial Communications | Build window: 28 December 2026-24 January 2027

Group	Quantity / specification	Reuse	Project role
Core process rig	PLC and selected Project 2 sensors/actuators	Shared	Provides live tags and credible process behaviour
Laptop/PC	1 x Ethernet-capable machine running Ignition and engineering tools	P1-P4	Use a dedicated project backup folder
Ethernet switch	1 x 5-port 24 VDC switch	P1-P4	Separate training LAN preferred
Patch cables	4 x tested Cat5e/Cat6 cables	P1-P4	Label both ends and keep one spare
RS-485 adapter	1 x isolated USB-to-RS-485 adapter	P3, P4	Isolation preferred for troubleshooting
Modbus RTU device	1 x documented 24 VDC remote I/O or sensor	P3, P4	Choose public register map and configurable address
Termination/bias kit	120 ohm termination plus documented biasing as required	P3, P4	Install only at correct bus locations
Optional second Ethernet device	Modbus TCP remote I/O, gateway or another controller	P3, P4	Not required if a simulator is used
Database	Local supported SQL database or Ignition historian configuration	P3, P4	Software item; back up schema/config

Network evidence to produce

- [] Address plan with device name, IP, subnet, protocol, port, node ID and ownership.
- [] Ignition device connection, tags, quality states, alarm priorities, trends and historical retrieval.
- [] Modbus RTU frame settings: baud, parity, stop bits, address, register, data type and byte/word order.
- [] Fault tests: unplugged cable, duplicate/wrong IP, stopped PLC, wrong serial settings, stale data and recovery.
- [] Packet capture or diagnostic screenshots with timestamps and a plain-language root-cause explanation.

SOFTWARE CHOICE

Ignition Maker Edition is appropriate for this personal, non-commercial learning portfolio and includes Perspective, OPC UA, historian and supported device drivers. A physical HMI is optional; the laptop can host the required interface.

Project 4 component deployment

Offshore / Remote Process Package Troubleshooting & Commissioning Simulation | Build window: 25 January-28 February 2027

Group	Quantity / specification	Origin	Project role
Complete bench	Reuse PLC, process rig, networking, indicators and software	P1-P3	Project 4 is a commissioning exercise, not a new shopping cycle
Fault-injection box	1 x labelled switch/patch box with protected open/short/substitution points	P4	Use only on extra-low-voltage circuits
Disconnect terminals	4-8 x knife/disconnect or test terminal positions	P4	Create repeatable, documented field faults
Spare sensor	1 x compatible PNP proximity/photoelectric sensor	P4	Substitution and proof-of-repair
Spare relay	1 x matching 24 VDC relay and base	P4	Coil/contact fault isolation
Spare fuses	Correct type/rating for every protected branch	P1-P4	Never uprate a fuse to clear a symptom
Portable backup	1 x USB drive or equivalent second storage location	P4	PLC, HMI, database, drawings and test pack
Calibration aids	Reuse DMM and 4-20 mA simulator; add leads/adapters if needed	P2-P4	Record as-found/as-left values
Optional DC buffer	24 VDC UPS/buffer sized for PLC/network load	P4	Optional brownout/ride-through demonstration

Minimum fault library

Layer	Faults to inject and diagnose
Electrical	Blown branch fuse, loose/open input, failed relay coil, missing 24 V common.
Instrumentation	4 mA underrange, 20 mA overrange, reversed loop polarity, incorrect scaling.
Control	Permissive missing, stale latch, bad mode ownership, restart not inhibited.
Network	Wrong IP/subnet, wrong Modbus node/ baud/parity, unplugged cable, stale SCADA data.
Process	Low-low level, no-flow with pump command, valve fail-to-open, duty pump unavailable.

PROCUREMENT GATE

Do not buy new Project 4 hardware until the fault matrix identifies a capability that the reused bench cannot safely simulate. The quality of evidence, diagnosis and restoration matters more than hardware volume.

Reuse matrix and staged buying plan

A filled cell shows the expected use of an item. P4 intentionally reuses nearly the whole bench.

Component group	P1	P2	P3	P4
PLC + analog module	X	X	X	X
24 VDC power/protection/terminals	X	X	X	X
Buttons/selectors/lamps/relays	X	X	X	X
Photoelectric/inductive/limit sensors	X	-	-	X
24 VDC geared motor/interface	X	-	-	X
0-10 V and 4-20 mA simulators	X	X	X	X
Tanks/pumps/tubing/valve	-	X	X	X
Process transmitters/flow device	-	X	X	X
Ethernet switch/cables	X	X	X	X
RS-485 adapter + Modbus device	-	-	X	X
Laptop: TIA + Ignition	X	X	X	X
Fault box, spares and test terminals	-	-	-	X

Staged acquisition gates

Purchase point	Buy / release	Gate before further spending
Before 5 Oct 2026	Core control bench + Project 1 sensors/motor	PLC powers up; every DI/DO proven with safe test loads; software connects; drawings match wiring.
Before 16 Nov 2026	Project 2 wet-process hardware	Dry simulation passes; one 4-20 mA loop correctly wired/scaled; leak test plan and containment ready.
Before 28 Dec 2026	Project 3 serial/network additions	Stable PLC Ethernet link; address plan complete; selected Modbus device has an official register map.
Before 25 Jan 2027	Project 4 fault-injection/spares only	Fault matrix reviewed; buy only missing capability; all injected faults remain at safe extra-low voltage.

BUDGET PRIORITY

Spend first on a compatible PLC/analog path, sound power distribution, safe wiring, measurement tools and documented sensors. Mechanical polish and optional vendor hardware come only after the control system is reliable and evidenced.

Compatibility, receiving and official references

Use this page before ordering and again when each item arrives.

Pre-order compatibility checklist

Check	What must match
PLC ecosystem	Exact CPU/module generation, TIA Portal support, expansion compatibility, firmware, memory card/licence needs.
Digital I/O	PNP/NPN convention, sourcing/sinking, output type, per-channel/group current, common terminals and inrush.
Analog I/O	0-10 V versus 4-20 mA, active/passive loop, two-/three-/four-wire transmitter, isolation and input impedance.
Power	Input supply, 24 V tolerance, steady/inrush load, branch fuse ratings, conductor size, earth and enclosure.
Networking	Ethernet/serial interface, connector, IP support, Modbus register map, termination, baud/parity and isolation.
Mechanical/process	Port/thread size, tubing material, wetted materials, pressure/temperature range, mounting and leak containment.

Receiving inspection

- [] Photograph label and condition; record supplier, exact article number, serial/firmware and datasheet revision.
- [] Check for counterfeit warning signs, damage, missing plugs/bases and mismatch between listing and product label.
- [] Do resistance/continuity and polarity checks unpowered; energize one protected branch at a time.
- [] Run a simple acceptance test, record measured values, and update the component register and spare list.

Official procurement references - verified 12 September 2026

Reference	Official URL
Siemens CPU 1212C	https://mall.industry.siemens.com/mall/en/gb/Catalog/Product/6ES72121AE400XB0
Siemens SM 1231 AI	https://mall.industry.siemens.com/goos/catalog/Pages/mmpdata.ashx?MLFB1=6ES7231-4HD32-0XB0(&=en
Mean Well HDR-100 series	https://www.meanwell.com/Upload/PDF/HDR-100/HDR-100-SPEC.PDF
Ignition Maker Edition	https://inductiveautomation.com/ignition/maker-edition
Ignition Maker manual	https://docs.inductiveautomation.com/docs/8.3/other-editions/ignition-maker-edition

FINAL PURCHASING CAUTION

Availability, regional article numbers, licences and product generations can change. Re-open the official datasheet and confirm the exact seller listing immediately before purchase. Prefer authorized distributors and keep invoices and datasheets with the portfolio evidence pack.